Vindhyawasini Prasad

+91-1332-28(5870) ◇ vprasad@ce.iitr.ac.in
Civil Engineering Department, Office #334A
Indian Institute of Technology Roorkee (India) - 247667

EDUCATION

University of Illinois at Urbana-Champaign *Aug 2019 - Dec 2024*
Doctor of Philosophy GPA: 3.89/4.0
Department of Civil and Environmental Engineering (Specialization Ecohydraulics)
Thesis: Development of Bubble Screen Deterrents for Controlling Movements of Invasive Carp Eggs and Larvae in Streams
Thesis committee: Prof. Rafael O. Tinoco (advisor), Prof. Marcelo Garcia, Prof. Cory Suski, Dr. P. Ryan Jackson and Dr. Daniel Zielinski

Clemson University *Aug 2017 - Aug 2019*
Master of Science GPA: 4.0/4.0
Glenn Department of Civil Engineering
Advisor: Prof. Kalyan Piratla

Indian Institute of Technology Roorkee (India) *July 2013 - May 2017*
Bachelor of Technology GPA: 7.743/10.0
Department of Civil Engineering

WORK EXPERIENCE

Indian Institute of Technology Roorkee *March 2026 - present*
Assistant Professor of Civil Engineering (Hydraulics Engineering specialization) *Roorkee, India*

· Experimental research in the broad area of ecohydraulics

Louisiana State University *Jan 2025 - March 2026*
Postdoctoral Researcher, Advisor: Dr. Muriel Brückner *Baton Rouge, LA, USA*

· Implemented a 2D hydrodynamic model ANUGA (Australian National University and Geoscience Australia) with vegetation life-cycle model to understand the feedback on inundation

AREAS OF INTEREST

Environmental Fluid Mechanics; Ecohydraulics; Flow-particle Interactions; and Multiphase Flows

PEER-REVIEWED JOURNAL PUBLICATIONS

Prasad, V., H. Doyle, Suski, C., Jackson, P., George, A., Chapman, D., Fischer, J., Tinoco, R. O. (2026), Controlling invasive carp ichthyoplankton dispersion using a streamwise-oriented bubble screen: A proof-of-concept validation in a laboratory flume, *Journal of Great Lakes Research*. doi.org/10.1016/j.jglr.2026.102784

Doyle, H.F., Stahlschmidt, B.H., Herndon, A.M., **Prasad, V.**, George, A.E., Fischer, J.R., Jackson, P.R., Suski, C.D., and Tinoco, R.O. (2026), Development and assessment of fluorescent-dyed, preserved invasive grass carp (*Ctenopharyngodon idella*) eggs as surrogates for live eggs in laboratory and field experiments, *River Research and Applications*, DOI: https://doi.org/10.1002/rra.70124.

Prasad, V., Andrade, J., Suski, C., Jackson, P., George, A., Chapman, D., Fischer, J., Tinoco, R. O. (2025), Efficacy of oblique bubble screen deterrents on redirecting live eggs and larvae of grass carp (*Ctenopharyngodon idella*) at different developmental stages, *Water Resources Research, 61, e2025WR039967.*, DOI: https://doi.org/10.1029/2025WR039967.

Prasad, V., Suski, C., Jackson, P., George, A., Chapman, D., Fischer, J., Tinoco, R. O. (2024), Limiting downstream dispersal of invasive carp egg surrogates using a laboratory-scale oblique bubble screen, *Journal of Ecohydraulics* DOI: 10.1080/24705357.2024.2332994

Momeni, A., Dharmawardena, H., **Prasad, V.**, Piratla, K., Venayagamoorthy, K. (2024), Modelling near-real-time multiphase cascading failures across interdependent power-water-gas infrastructures, *Sustainable and Resilient Infrastructure* DOI: 10.1080/15732479.2024.2312267

Yadav, V., Sherly, M.A., Ranjan, P., **Prasad, V.**, Tinoco, R.O., Laurent, A. (2022), Risk of plastics losses to the environment from Indian landfills, *Resources, Conservation and Recycling 187*(106610) DOI: 10.1016/j.resconrec.2022.106610

Prasad, V., K. Piratla (2021), Optimal Rehabilitation of Gas Distribution Pipelines in Liquefiable Soils Considering Seismic Reliability, *ASCE Journal of Pipeline Systems Engineering and Practices 12*(3) DOI:10.1061/(ASCE)PS.1949-1204.0000545

UNDER-REVIEW JOURNAL PUBLICATIONS

Prasad, V., H. Doyle, Suski, C., Jackson, P., George, A., Chapman, D., Fischer, J., Tinoco, R. O., Development of high-fidelity surrogates of invasive carp eggs for laboratory experimentation: Comparison with real eggs through oblique bubble screen deterrents, *(under USGS internal peer-review)*

Doyle, H.F., **Prasad, V.**, Stahlschmidt, B.H., Herndon, A., George, A.E., Fischer, J.R., Jackson, P.R., Suski, C.D., and Tinoco, R.O., Using fluorescent-dyed preserved grass carp *(Ctenopharyngodon idella)* eggs for particle tracking velocimetry in laboratory experiments , (reviewed and submitted to *Journal of Ecohydraulics)*

IN PREPARATION JOURNAL PUBLICATIONS

Farrow, C.,**Prasad, V.**, H. Doyle, Suski, C., Jackson, P., George, A., J. Ackerman, Tinoco, R. O. (In preparation), Alginate beads as a environmentally friendly surrogates for invasive carp eggs in laboratory and field applications, *(in collaboration with researchers at the University of Guelph, Canada)*

Prasad, V., Tinoco, R. O. (In preparation), Hydrodynamics within and around root canopies of floating vegetation patches, *(Part of PhD research)*

CONFERENCE PRESENTATIONS/ POSTERS

Prasad, V., R. Jackson, Fischer, J., George, A., Chapman, D., H. Doyle, R.O. Tinoco (2024), Bubble screens as deterrents to facilitate the capture of downstream drifting eggs and larvae of invasive grass carp (*Ctenopharyngodon idella)* in streams, Oral Presentation at the *FAmerican Geophysical Union Fall Meeting 2024, Washington D.C., USA, Dec 9-15, 2024*

Prasad, V., R. Jackson, Fischer, J., George, A., Chapman, D., H. Doyle, R.O. Tinoco (2024), Assessment of bubble screens as a dispersal barrier to control the movement of drifting invasive grass carp (*Ctenopharyngodon idella)* eggs, Oral Presentation at the *15th International Symposium on Ecohydraulics and Fish Passage, Quebec City, Canada, May 5-9, 2024*

Prasad, V., R. Jackson, Fischer, J., George, A., Chapman, D., H. Doyle, R.O. Tinoco (2024), Oblique bubble screens as two-way multi-life stage dispersal barriers, Presented the research updates at *Great Rivers Cooperative Ecosystem Studies Unit (CESU) annual meeting, March 12, 2024* (virtual)

Doyle, Henry, **Prasad, V.**, Suski, C., P.R. Jackson, George, A., B. Stahlschmidt, Chapman, D., J.R. Fischer, and R. O. Tinoco (2023), Developing an oblique bubble screen to redirect drifting eggs in streams: addressing challenges of using live eggs in laboratory tests, Oral presentation at *13th Symposium on River, Coastal and Estuarine Morphodynamics, Urbana, Illinois, Sept 25-28, 2023*

Prasad, V., Tinoco, R. O.(2023), Hydrodynamics within and around floating vegetation patches, Poster presented at *13th Symposium on River, Coastal and Estuarine Morphodynamics, Urbana, Illinois, Sept 25-28, 2023*

Prasad, V., Tinoco, R. O.(2023), Hydrodynamics within and around floating vegetation patches, Poster presented at *9th Young Coastal Scientists and Engineers Conference - Americas, Madison, Wisconsin, Aug 10-12, 2023*

Prasad, V., Tinoco, R. O.(2022), Assessing the Efficacy of Oblique Bubble Screens for Control of Aquatic Invasive Species, Extended abstract presented at *39th IAHR World Congress, Granada, Spain, 19-24 June 2022* [Peer-reviewed]

Prasad, V., Tinoco, R. O.(2021), Hydrodynamics in open channel flow with floating vegetation: a comparison of simplified geometry and complex root canopies, Extended abstract presented in *5th International Symposium of Shallow Flows, Nanjing, China, 23-25 October 2021*

Prasad, V., Tinoco, R. O.(2020), From Surface to Substrate: Impact of Floating Vegetation Root-Canopies on Turbulence and Hydrodynamics in Streams, Poster presented in *American Geophysical Union Fall Meeting 2020*

Prasad, V., K. Piratla, B. Caswell (2020), Seismic Reliability Evaluation of City-level Gas Distribution Networks Using Flow-based Simulation Modeling, Full paper presented in *ASCE Construction Research Congress-2020, Tempe, AZ.* [Peer-reviewed]

Momeni, A., **Prasad, V.**, K. Piratla, H. Dharmawardena, V. Kumar (2018), Modeling Interdependent Power, Water, and Gas Infrastructures, Full paper presented in *Power Systems Conference 2018, Clemson University* [Peer-reviewed]

RESEARCH EXPERIENCE

Louisiana State University *Jan 2025 - ongoing*
Postdoctoral Researcher, Advisor: Dr. Muriel Bruckner *Baton Rouge, LA, USA*

· Numerical modeling of the evolution of Wax Lake Delta, a part of the Mississippi River Delta system in coastal Louisiana

- Implementing 2D hydrodynamic model ANUGA (Australian National University and Geoscience Australia) with vegetation life-cycle model to understand the feedback on sediment accretion

Oblique bubble screen to control the passage of aquatic invasive species *Aug 2021 - July 2024*
Part of PhD dissertation, Advisor: Prof. Rafael Tinoco *UIUC*

- Using bubble screens to deter the movement of grass carp, an invasive fish species in Midwest region in its different life stages - eggs, larvae and young fish
- Different configurations of the bubble screen were developed and tested in a laboratory flume
- Detailed hydrodynamics around the bubble screen was acquired using ADV and surface PIV. The data helped us predict the movement of fish eggs for different configurations of the bubble screens.

Open channel flow with floating vegetation *Aug 2019 - Aug 2022*
Part of PhD dissertation, Advisor: Prof. Rafael Tinoco *UIUC*

- Experimental study to investigate the impacts of suspended canopies on hydrodynamics and sediment transport in open channel flow
- Different surrogates were used to study the impact of the morphology of the canopies. PIV data showed different scales of turbulence present within and underneath the suspended root-canopies.

Optimal Rehabilitation of Gas Distribution Pipelines in Liquefiable Soils Considering Seismic Reliability *Aug 2017 - Aug 2019*
Master's thesis project, Advisor: Prof. Kalyan Piratla *Clemson University*

- Developed framework for optimal rehabilitation of city-level gas distribution network for regions prone to soil-liquefaction using numerical models
- Integrated numerical gas-flow models and pipeline fragility models subject to historical seismic event to predict the disruption at the network level
- Genetic algorithm optimization was used in optimal design and rehabilitation planning of the network

Data driven co-simulation models of interdependent infrastructure systems for adaptive control [Abstract-Link] *Aug 2017 - Aug 2019*
Graduate research project, Advisor: Prof. Kalyan Piratla *Clemson University*

- Developed numerical models for natural gas transmission and distribution network from scratch. The model was co-simulated with power systems and water distribution network at small time steps to study the cascading effects among Power-Water-Gas infrastructures.
- The modeling framework is demonstrated using real-world infrastructure data from Charleston region in South Carolina, USA.

Effects of downward seepage on boundary layer *May 2016 - July 2016*
Summer Research Intern, Advisor: Prof. Bimlesh Kumar *IIT Guwahati, India*

- Assisted in conducting experiments to investigate the effects of downward seepage on boundary layer in an alluvial rectangular flume
- Also developed Matlab code for efficient noise-removal from velocity time series recorded using ADV (acoustic Doppler velocimeter). Results were presented at a national conference later.

TEACHING EXPERIENCE

IIT Roorkee *Fall 2026 onward*
jCourse Instructor *Roorkee, India*

- Taught CEC533 Advanced Fluid Mechanics (graduate level core course)

University of Illinois at Urbana-Champaign *Fall 2023 & Spring 2021*
Teaching Assistant *Urbana, IL*

· Teaching assistant for CEE331 - Fluid Mechanics in Natural and Built Environments, a newly developed course in CEE department
· Co-developed and conducted lab sessions for CEE331
· Teaching assistant for CEE350-Water Resources Engineering (Spring 2021)

Clemson University *Aug 2017 - Aug 2019*
Graduate Research Assistant & Teaching Assistant *Clemson,SC*

· Teaching assistant for courses 'Construction Engineering and Management', 'Economic Evaluation of Projects' and 'Construction Equipment Selection and Maintenance'

RELEVANT GRADUATE-LEVEL COURSES

Environmental Fluid Mechanics	Experimental Fluid Mechanics
Sediment Transport	River Morphodynamics
Turbulence	Groundwater Modeling
Mixing in Environmental Flows	Environmental Hydrodynamics

TECHNICAL STRENGTHS

Experimental techniques	Particle image velocimetry (PIV), acoustic Doppler velocimetry (ADV), particle tracking velocimetry (PTV), laser induced fluorescence (LIF)
Computer Languages	Matlab, Python, R, C/C++
Software Packages	Delft3D, ANUGA, ANSYS Fluent, QGIS, ArcGIS

FELLOWSHIPS & AWARDS

Recipient of *Conference Presentation Travel Award Spring 2024* (the Graduate College, UIUC)

2nd best poster award at *Young Coastal Scientists and Engineers Conference - Americas, Madison, Wisconsin, Aug 2023*

Ranked as an excellent TA (outstanding category: 4.59/5.0) in course evaluation of CEE331 in Fall 2023

Recipient of *Mavis Future Faculty Fellowship* for the academic year 2022-23 (by Grainger College of Engineering at UIUC)

Recipient of *Ravindra K. and Kavita Kinra Fellowship* for the academic year 2019-20 at UIUC

Recipient of merit-cum-means scholarship awarded by the Government of India for the years 2014-17

EXTRA-CURRICULAR ENGAGEMENTS

Part of the local organizing committee of 13th Symposium on River, Coastal and Estuarine Morphodynamics (RCEM) held at UIUC, Urbana, IL (Sept 2023).

Treasurer at International Water Resources Association (IWRA) students chapter at UIUC (2021-22)

Led and volunteered in multiple exhibits at Ven Te Chow Hydrosystems Laboratory during Engineering Open House (EOH) organized by Grainger College of Engineering at UIUC (2020-2024)

Vice-president of Indian Graduate Students Association at UIUC (2021-22)

Liaison for EEL-GEICSW (Global Engineering Innovation Center for Secure Water) website

REFERENCES

Available on request